Labsheet 7

1. 00_setup.sql

```
CREATE DATABASE college_db;
USE college_db;

CREATE TABLE departments (
    dept_id   INT PRIMARY KEY,
    dept_name VARCHAR(50) NOT NULL UNIQUE
);

CREATE TABLE students (
    student_id INT PRIMARY KEY,
    name       VARCHAR(50) NOT NULL,
    age        INT CHECK (age >= 17),
    city       VARCHAR(50),
    dept_id    INT,
    FOREIGN KEY (dept_id) REFERENCES departments(dept_id)
);

CREATE TABLE courses (
    course_id INT PRIMARY KEY,
    title     VARCHAR(80) NOT NULL,
    credits   INT DEFAULT 3,
    dept_id   INT,
    FOREIGN KEY (dept_id) REFERENCES departments(dept_id)
);

CREATE TABLE enrollments (
    student_id INT,
    course_id  INT,
    marks      INT,
    PRIMARY KEY (student_id, course_id),
    FOREIGN KEY (student_id) REFERENCES students(student_id),
    FOREIGN KEY (course_id)  REFERENCES courses(course_id)
);

INSERT INTO departments VALUES (1,'CSE'),(2,'ECE'),(3,'Mechanical'),(4,'Civil');

INSERT INTO students VALUES
(101,'Aarav',20,'Roorkee',1),
(102,'Diya',21,'Dehradun',1),
(103,'Kabir',19,'Haridwar',2),
(104,'Meera',22,'Roorkee',1),
(105,'Rohan',20,'Delhi',3),
(106,'Sana',21,'Dehradun',2),
(107,'Vikram',23,'Jaipur',NULL),
(108,'Isha',19,'Roorkee',2);

INSERT INTO courses VALUES
(201,'Data Structures',4,1),
(202,'Operating Systems',4,1),
(203,'Digital Electronics',3,2),
(204,'Thermodynamics',3,3),
(205,'Machine Learning',4,1);

INSERT INTO enrollments VALUES
(101,201,88),(101,202,76),(101,205,91),
(102,201,67),(102,205,72),
(103,203,81),
(104,201,95),(104,202,89),
(105,204,58),
(106,203,74),
(108,203,66);

SELECT * FROM students;
```

2. 01_part_a_ddl.sql

```
-- A1: Create faculty table
CREATE TABLE faculty (
    faculty_id INT PRIMARY KEY,
    name       VARCHAR(50) NOT NULL,
    email      VARCHAR(100) UNIQUE,
    salary     DECIMAL(10,2),
    dept_id    INT,
    FOREIGN KEY (dept_id) REFERENCES departments(dept_id)
);

-- A2: Add phone column
ALTER TABLE faculty ADD phone VARCHAR(15);

-- A3: Rename column phone to contact_no
ALTER TABLE faculty RENAME COLUMN phone TO contact_no;

-- A4: Drop column contact_no
ALTER TABLE faculty DROP COLUMN contact_no;

-- A5: Drop faculty table
DROP TABLE faculty;
```

3. 02_part_b_dml.sql

```
-- B1: Insert new student
INSERT INTO students VALUES (109, 'Neha', 20, 'Roorkee', 1);

-- B2: Insert without city
INSERT INTO students (student_id, name, age, dept_id) VALUES (110, 'Rahul', 21, 2);

-- B3: Update city of student 105
UPDATE students SET city = 'Roorkee' WHERE student_id = 105;

-- B4: Increase marks by 5 for course 203
UPDATE enrollments SET marks = marks + 5 WHERE course_id = 203;

-- B5: Delete student 109 and 110
DELETE FROM students WHERE student_id IN (109, 110);

-- B6: Age < 17 test (Will throw Check Constraint Error)
-- INSERT INTO students VALUES (111, 'Amit', 15, 'Delhi', 1);
```

4. 03_part_c_select.sql

```
-- C1
SELECT * FROM students;

-- C2
SELECT name, city FROM students;

-- C3
SELECT * FROM students WHERE city = 'Roorkee';

-- C4
SELECT * FROM students WHERE age BETWEEN 20 AND 22;

-- C5
SELECT * FROM students WHERE city IN ('Roorkee', 'Dehradun');

-- C6
SELECT * FROM students WHERE name LIKE 'A%';

-- C7
```

```
SELECT * FROM students WHERE dept_id IS NULL;

-- C8
SELECT DISTINCT city FROM students;

-- C9
SELECT * FROM students ORDER BY age DESC, name ASC;

-- C10
SELECT * FROM students ORDER BY age DESC LIMIT 3;

-- C11
SELECT name AS "Student Name", age AS "Age" FROM students;
```

5. 04_part_d_aggregation.sql

```
-- D1
SELECT COUNT(*) AS total_students FROM students;

-- D2
SELECT AVG(marks) AS avg_marks, MIN(marks) AS min_marks, MAX(marks) AS max_marks FROM enrollments;

-- D3
SELECT city, COUNT(*) AS student_count FROM students GROUP BY city;

-- D4
SELECT course_id, AVG(marks) AS avg_marks FROM enrollments GROUP BY course_id;

-- D5
SELECT city, COUNT(*) FROM students GROUP BY city HAVING COUNT(*) > 1;

-- D6
SELECT dept_id, SUM(credits) AS total_credits FROM courses GROUP BY dept_id;

-- D7
SELECT UPPER(name) AS upper_name, LENGTH(name) AS name_length FROM students;
```

6. 05_part_e_joins.sql

```
-- E1
SELECT s.name, d.dept_name
FROM students s
JOIN departments d ON s.dept_id = d.dept_id;

-- E2
SELECT s.name, d.dept_name
FROM students s
LEFT JOIN departments d ON s.dept_id = d.dept_id;

-- E3
SELECT d.dept_name, s.name
FROM departments d
LEFT JOIN students s ON d.dept_id = s.dept_id;

-- E4
SELECT s.name, c.title, e.marks
FROM students s
JOIN enrollments e ON s.student_id = e.student_id
JOIN courses c ON e.course_id = c.course_id;

-- E5
SELECT s.name
FROM students s
JOIN enrollments e ON s.student_id = e.student_id
JOIN courses c ON e.course_id = c.course_id
WHERE c.title = 'Data Structures';
```

```sql
-- E6
SELECT s.name
FROM students s
LEFT JOIN enrollments e ON s.student_id = e.student_id
WHERE e.course_id IS NULL;

-- E7
SELECT c.title, COUNT(e.student_id) AS enrolled_students
FROM courses c
LEFT JOIN enrollments e ON c.course_id = e.course_id
GROUP BY c.course_id, c.title;

-- E8
SELECT d.dept_name, COUNT(s.student_id) AS total_students, AVG(s.age) AS avg_age
FROM departments d
LEFT JOIN students s ON d.dept_id = s.dept_id
GROUP BY d.dept_id, d.dept_name;
```

7. 06_part_f_subqueries.sql

```sql
-- F1
SELECT * FROM students
WHERE age > (SELECT AVG(age) FROM students);

-- F2
SELECT DISTINCT s.*
FROM students s
JOIN enrollments e ON s.student_id = e.student_id
WHERE e.marks = (SELECT MAX(marks) FROM enrollments);

-- F3
SELECT * FROM courses
WHERE course_id IN (SELECT DISTINCT course_id FROM enrollments);

-- F4
SELECT student_id
FROM enrollments
GROUP BY student_id
HAVING COUNT(course_id) > 2;

-- F5
SELECT dept_id
FROM students
WHERE dept_id IS NOT NULL
GROUP BY dept_id
ORDER BY COUNT(*) DESC LIMIT 1;

-- F6
SELECT e1.student_id, e1.course_id, e1.marks
FROM enrollments e1
WHERE e1.marks > (
    SELECT AVG(e2.marks)
    FROM enrollments e2
    WHERE e2.course_id = e1.course_id
);
```

8. 07_part_g_advanced.sql

```sql
-- G1
CREATE INDEX idx_student_city ON students(city);
EXPLAIN SELECT * FROM students WHERE city = 'Roorkee';

-- G2
CREATE VIEW student_marks AS
SELECT s.name AS student_name, c.title AS course_title, e.marks
FROM students s
```

```
JOIN enrollments e ON s.student_id = e.student_id
JOIN courses c ON e.course_id = c.course_id;

-- G3
SELECT * FROM student_marks WHERE marks > 80;

-- G4
DROP VIEW student_marks;
DROP INDEX idx_student_city ON students;

-- G5
START TRANSACTION;
UPDATE enrollments SET marks = 100 WHERE student_id = 101 AND course_id = 201;
ROLLBACK;

-- G6
START TRANSACTION;
UPDATE enrollments SET marks = 100 WHERE student_id = 101 AND course_id = 201;
COMMIT;
```

9. 08_part_h_challenge.sql

```
-- H1: Rank List
SELECT student_id, SUM(marks) AS total_marks,
       RANK() OVER (ORDER BY SUM(marks) DESC) AS rank_num,
       DENSE_RANK() OVER (ORDER BY SUM(marks) DESC) AS dense_rank_num
FROM enrollments
GROUP BY student_id;

-- H2: Grade Assignment
SELECT student_id, course_id, marks,
       CASE
           WHEN marks >= 90 THEN 'A'
           WHEN marks >= 75 THEN 'B'
           WHEN marks >= 60 THEN 'C'
           ELSE 'D'
       END AS grade
FROM enrollments;

-- H3: Self Join for Same City
SELECT s1.name AS student1, s2.name AS student2, s1.city
FROM students s1
JOIN students s2 ON s1.city = s2.city AND s1.student_id < s2.student_id;

-- H4: Second Highest Marks
SELECT MAX(marks) AS second_highest_marks
FROM enrollments
WHERE marks < (SELECT MAX(marks) FROM enrollments);
```

Output

1. 00_setup.sql

SELECT * FROM students;

```
+------------+--------+-----+----------+---------+
| student_id | name   | age | city     | dept_id |
+------------+--------+-----+----------+---------+
| 101        | Aarav  | 20  | Roorkee  | 1       |
| 102        | Diya   | 21  | Dehradun | 1       |
| 103        | Kabir  | 19  | Haridwar | 2       |
| 104        | Meera  | 22  | Roorkee  | 1       |
| 105        | Rohan  | 20  | Delhi    | 3       |
| 106        | Sana   | 21  | Dehradun | 2       |
| 107        | Vikram | 23  | Jaipur   | NULL    |
| 108        | Isha   | 19  | Roorkee  | 2       |
+------------+--------+-----+----------+---------+
8 rows in set
```

2. 01_part_a_ddl.sql

A1

```
Query OK, 0 rows affected
```

A2

```
Query OK, 0 rows affected
```

A3

```
Query OK, 0 rows affected
```

A4

```
Query OK, 0 rows affected
```

A5

```
Query OK, 0 rows affected
```

3. 02_part_b_dml.sql

B1

```
Query OK, 1 row affected
```

B2

```
Query OK, 1 row affected
```

B3

```
Query OK, 1 row affected
```

B4

```
Query OK, 3 rows affected
```

B5

```
Query OK, 2 rows affected
```

B6

```
ERROR 3819 (HY000): Check constraint 'students_chk_1' is violated.
```

4. 03_part_c_select.sql

C1

student_id	name	age	city	dept_id
101	Aarav	20	Roorkee	1
102	Diya	21	Dehradun	1
103	Kabir	19	Haridwar	2
104	Meera	22	Roorkee	1
105	Rohan	20	Roorkee	3
106	Sana	21	Dehradun	2
107	Vikram	23	Jaipur	NULL
108	Isha	19	Roorkee	2

8 rows in set

C2

name	city
Aarav	Roorkee
Diya	Dehradun
Kabir	Haridwar
Meera	Roorkee
Rohan	Roorkee
Sana	Dehradun
Vikram	Jaipur
Isha	Roorkee

8 rows in set

C3

student_id	name	age	city	dept_id
101	Aarav	20	Roorkee	1
104	Meera	22	Roorkee	1
105	Rohan	20	Roorkee	3
108	Isha	19	Roorkee	2

4 rows in set

C4

student_id	name	age	city	dept_id
101	Aarav	20	Roorkee	1
102	Diya	21	Dehradun	1
104	Meera	22	Roorkee	1
105	Rohan	20	Roorkee	3
106	Sana	21	Dehradun	2

5 rows in set

C5

student_id	name	age	city	dept_id
101	Aarav	20	Roorkee	1
102	Diya	21	Dehradun	1
104	Meera	22	Roorkee	1
105	Rohan	20	Roorkee	3
106	Sana	21	Dehradun	2
108	Isha	19	Roorkee	2

6 rows in set

C6

```
+------------+-------+-----+---------+---------+
| student_id | name  | age | city    | dept_id |
+------------+-------+-----+---------+---------+
| 101        | Aarav | 20  | Roorkee | 1       |
+------------+-------+-----+---------+---------+
1 row in set
```

C7

```
+------------+--------+-----+--------+---------+
| student_id | name   | age | city   | dept_id |
+------------+--------+-----+--------+---------+
| 107        | Vikram | 23  | Jaipur | NULL    |
+------------+--------+-----+--------+---------+
1 row in set
```

C8

```
+----------+
| city     |
+----------+
| Roorkee  |
| Dehradun |
| Haridwar |
| Jaipur   |
+----------+
4 rows in set
```

C9

```
+------------+--------+-----+----------+---------+
| student_id | name   | age | city     | dept_id |
+------------+--------+-----+----------+---------+
| 107        | Vikram | 23  | Jaipur   | NULL    |
| 104        | Meera  | 22  | Roorkee  | 1       |
| 102        | Diya   | 21  | Dehradun | 1       |
| 106        | Sana   | 21  | Dehradun | 2       |
| 101        | Aarav  | 20  | Roorkee  | 1       |
| 105        | Rohan  | 20  | Roorkee  | 3       |
| 108        | Isha   | 19  | Roorkee  | 2       |
| 103        | Kabir  | 19  | Haridwar | 2       |
+------------+--------+-----+----------+---------+
8 rows in set
```

C10

```
+------------+--------+-----+----------+---------+
| student_id | name   | age | city     | dept_id |
+------------+--------+-----+----------+---------+
| 107        | Vikram | 23  | Jaipur   | NULL    |
| 104        | Meera  | 22  | Roorkee  | 1       |
| 102        | Diya   | 21  | Dehradun | 1       |
+------------+--------+-----+----------+---------+
3 rows in set
```

C11

```
+--------------+-----+
| Student Name | Age |
+--------------+-----+
| Aarav        | 20  |
| Diya         | 21  |
| Kabir        | 19  |
| Meera        | 22  |
| Rohan        | 20  |
| Sana         | 21  |
| Vikram       | 23  |
| Isha         | 19  |
+--------------+-----+
8 rows in set
```

5. 04_part_d_aggregation.sql

D1

```
+----------------+
| total_students |
+----------------+
| 8              |
+----------------+
1 row in set
```

D2

```
+-----------+-----------+-----------+
| avg_marks | min_marks | max_marks |
+-----------+-----------+-----------+
| 79.2727   | 58        | 95        |
+-----------+-----------+-----------+
1 row in set
```

D3

```
+----------+---------------+
| city     | student_count |
+----------+---------------+
| Dehradun | 2             |
| Haridwar | 1             |
| Jaipur   | 1             |
| Roorkee  | 4             |
+----------+---------------+
4 rows in set
```

D4

```
+-----------+-----------+
| course_id | avg_marks |
+-----------+-----------+
| 201       | 83.3333   |
| 202       | 82.5000   |
| 203       | 78.6667   |
| 204       | 58.0000   |
| 205       | 81.5000   |
+-----------+-----------+
5 rows in set
```

D5

```
+----------+----------+
| city     | COUNT(*) |
+----------+----------+
| Dehradun | 2        |
| Roorkee  | 4        |
+----------+----------+
2 rows in set
```

D6

```
+---------+---------------+
| dept_id | total_credits |
+---------+---------------+
| 1       | 12            |
| 2       | 3             |
| 3       | 3             |
+---------+---------------+
3 rows in set
```

D7

```
+------------+-------------+
| upper_name | name_length |
+------------+-------------+
| AARAV      | 5           |
| DIYA       | 4           |
| KABIR      | 5           |
| MEERA      | 5           |
| ROHAN      | 5           |
| SANA       | 4           |
| VIKRAM     | 6           |
| ISHA       | 4           |
+------------+-------------+
8 rows in set
```

6. 05_part_e_joins.sql

E1

```
+-------+------------+
| name  | dept_name  |
+-------+------------+
| Aarav | CSE        |
| Diya  | CSE        |
| Kabir | ECE        |
| Meera | CSE        |
| Rohan | Mechanical |
| Sana  | ECE        |
| Isha  | ECE        |
+-------+------------+
7 rows in set
```

E2

```
+--------+------------+
| name   | dept_name  |
+--------+------------+
| Aarav  | CSE        |
| Diya   | CSE        |
| Kabir  | ECE        |
| Meera  | CSE        |
| Rohan  | Mechanical |
| Sana   | ECE        |
| Vikram | NULL       |
| Isha   | ECE        |
+--------+------------+
8 rows in set
```

E3

```
+------------+-------+
| dept_name  | name  |
+------------+-------+
| CSE        | Aarav |
| CSE        | Diya  |
| CSE        | Meera |
| ECE        | Kabir |
| ECE        | Sana  |
| ECE        | Isha  |
| Mechanical | Rohan |
| Civil      | NULL  |
+------------+-------+
8 rows in set
```

E4

```
+-------+--------------------+-------+
| name  | title              | marks |
+-------+--------------------+-------+
| Aarav | Data Structures    | 88    |
| Aarav | Operating Systems  | 76    |
| Aarav | Machine Learning   | 91    |
| Diya  | Data Structures    | 67    |
| Diya  | Machine Learning   | 72    |
| Kabir | Digital Electronics| 86    |
| Meera | Data Structures    | 95    |
| Meera | Operating Systems  | 89    |
| Rohan | Thermodynamics     | 58    |
| Sana  | Digital Electronics| 79    |
| Isha  | Digital Electronics| 71    |
+-------+--------------------+-------+
11 rows in set
```

E5

```
+-------+
| name  |
+-------+
| Aarav |
| Diya  |
| Meera |
+-------+
3 rows in set
```

E6

```
+--------+
| name   |
+--------+
| Vikram |
+--------+
1 row in set
```

E7

```
+--------------------+-------------------+
| title              | enrolled_students |
+--------------------+-------------------+
| Data Structures    | 3                 |
| Operating Systems  | 2                 |
| Digital Electronics| 3                 |
| Thermodynamics     | 1                 |
| Machine Learning   | 2                 |
+--------------------+-------------------+
5 rows in set
```

E8

```
+------------+----------------+---------+
| dept_name  | total_students | avg_age |
+------------+----------------+---------+
| CSE        | 3              | 21.0000 |
| Civil      | 0              | NULL    |
| ECE        | 3              | 19.6667 |
| Mechanical | 1              | 20.0000 |
+------------+----------------+---------+
4 rows in set
```

7. 06_part_f_subqueries.sql

F1

```
+------------+--------+-----+----------+---------+
| student_id | name   | age | city     | dept_id |
+------------+--------+-----+----------+---------+
| 102        | Diya   | 21  | Dehradun | 1       |
| 104        | Meera  | 22  | Roorkee  | 1       |
| 106        | Sana   | 21  | Dehradun | 2       |
| 107        | Vikram | 23  | Jaipur   | NULL    |
+------------+--------+-----+----------+---------+
4 rows in set
```

F2

```
+------------+-------+-----+---------+---------+
| student_id | name  | age | city    | dept_id |
+------------+-------+-----+---------+---------+
| 104        | Meera | 22  | Roorkee | 1       |
+------------+-------+-----+---------+---------+
1 row in set
```

F3

```
+-----------+--------------------+---------+---------+
| course_id | title              | credits | dept_id |
+-----------+--------------------+---------+---------+
| 201       | Data Structures    | 4       | 1       |
| 202       | Operating Systems  | 4       | 1       |
| 203       | Digital Electronics | 3      | 2       |
| 204       | Thermodynamics     | 3       | 3       |
| 205       | Machine Learning   | 4       | 1       |
+-----------+--------------------+---------+---------+
5 rows in set
```

F4

```
+------------+
| student_id |
+------------+
| 101        |
+------------+
1 row in set
```

F5

```
+---------+
| dept_id |
+---------+
| 1       |
+---------+
1 row in set
```

F6

```
+------------+-----------+-------+
| student_id | course_id | marks |
+------------+-----------+-------+
| 101        | 201       | 88    |
| 101        | 205       | 91    |
| 103        | 203       | 86    |
| 104        | 201       | 95    |
| 104        | 202       | 89    |
| 106        | 203       | 79    |
+------------+-----------+-------+
6 rows in set
```

8. 07_part_g_advanced.sql

G1

```
Query OK, 0 rows affected
```

G2

```
Query OK, 0 rows affected
```

G3

```
+--------------+--------------------+-------+
| student_name | course_title       | marks |
+--------------+--------------------+-------+
| Aarav        | Data Structures    | 88    |
| Aarav        | Machine Learning   | 91    |
| Kabir        | Digital Electronics | 86   |
| Meera        | Data Structures    | 95    |
| Meera        | Operating Systems  | 89    |
+--------------+--------------------+-------+
5 rows in set
```

G4

```
Query OK, 0 rows affected
Query OK, 0 rows affected
```

G5

```
Query OK, 0 rows affected
Query OK, 1 row affected
Query OK, 0 rows affected
```

G6

```
Query OK, 0 rows affected
Query OK, 1 row affected
Query OK, 0 rows affected
```

9. 08_part_h_challenge.sql

H1

```
+------------+-------------+----------+----------------+
| student_id | total_marks | rank_num | dense_rank_num |
+------------+-------------+----------+----------------+
| 101        | 267         | 1        | 1              |
| 104        | 184         | 2        | 2              |
| 102        | 139         | 3        | 3              |
| 103        | 86          | 4        | 4              |
| 106        | 79          | 5        | 5              |
| 108        | 71          | 6        | 6              |
| 105        | 58          | 7        | 7              |
+------------+-------------+----------+----------------+
7 rows in set
```

H2

```
+------------+-----------+-------+-------+
| student_id | course_id | marks | grade |
+------------+-----------+-------+-------+
| 101        | 201       | 100   | A     |
| 101        | 202       | 76    | B     |
| 101        | 205       | 91    | A     |
| 102        | 201       | 67    | C     |
| 102        | 205       | 72    | C     |
| 103        | 203       | 86    | B     |
| 104        | 201       | 95    | A     |
| 104        | 202       | 89    | B     |
| 105        | 204       | 58    | D     |
| 106        | 203       | 79    | B     |
| 108        | 203       | 71    | C     |
+------------+-----------+-------+-------+
11 rows in set
```

H3

```
+----------+----------+----------+
| student1 | student2 | city     |
+----------+----------+----------+
| Aarav    | Meera    | Roorkee  |
| Aarav    | Rohan    | Roorkee  |
| Aarav    | Isha     | Roorkee  |
| Diya     | Sana     | Dehradun |
| Meera    | Rohan    | Roorkee  |
| Meera    | Isha     | Roorkee  |
| Rohan    | Isha     | Roorkee  |
+----------+----------+----------+
7 rows in set
```

H4

```
+----------------------+
| second_highest_marks |
+----------------------+
| 95                   |
+----------------------+
1 row in set
```